Rhayader: Cortical Column Critters — Design Notes

Overview

Next step after the Data.SDR work: set up Izhikevich spiking neurons organised as cortical columns, interpret spikes without relying on rate coding, and explore treating NEAT critters as columns that coevolve in teams.

Spike Interpretation

Rate coding is avoided. Candidate schemes:

- **Latency coding** — time to first spike carries the value.
- **Rank-order coding** — the order in which neurons fire matters.
- **Phase coding** — spike timing relative to a background oscillation.
- **Polychronous groups** — precise time-locked firing patterns shaped by axonal delays and STDP.

Bridge to SDRs: neurons spiking within a time window become active bits; rank or latency can add weighting.

Column Structure

- Excitatory regular-spiking neurons plus fast-spiking inhibitory neurons, roughly 4:1.
- First excitatory spikes drive the inhibitory cells, which suppress the rest of the column (winner-take-most).

Inhibition Strategy

1. Start with inhibition **within each column only** — simple and highly parallel.
2. Later add weak, **distance-weighted cross-column inhibition** as a tunable parameter; raise it if neighbouring columns duplicate each other.

Critters as Columns

Idea under consideration: each NEAT critter *is* a column. The original Ruby NEAT had critters competing (winner mates) plus speciation; this shifts evolution toward cooperating parts.

Teams and Cooperative Coevolution

- Groups of critters (teams) evolve together, in the spirit of SANE, ESP and CoSyNE.
- Each critter is scored by the performance of the teams it joins.
- Credit assignment is the hard part. Mitigations: evaluate each critter in several random teams; keep an elite best team and test newcomers against it.
- Iteration cost is uncertain: smaller search spaces may cut evaluations, but noisy team trials need more trials per critter. Benchmark against plain NEAT on a simple task.

Differentiation Within a Team

All four mechanisms, layered in this order:

1. **Random receptive fields** — each critter samples random places on the input SDR (like HTM potential pools); the sampling map may become part of the genome so evolution refines it.
2. **Cross-column inhibition** — the tunable dial above.
3. **Per-role subpopulations** — each team slot evolves in its own population.
4. **Diversity reward** — a tunable fitness weight penalising redundant outputs.

Neuron Coordinates and Drift

Every neuron gets a spatial coordinate stored in its gene, and positions drift over evolution.

- **Delays from distance**: distance divided by conduction speed, rounded to the timestep, feeds a per-delay ring buffer of pending spikes. This supports polychronous groups.
- **Geometry-driven inhibition and wiring**: cross-column inhibition and connection probability fall off with distance. Mutations can favour nearby targets.
- **Columns as regions**: a critter is a spatial cluster and a team is neighbouring clusters. Receptive fields can follow position, like a cortical map.
- **Spatial indexing**: a uniform grid or spatial hash handles neighbour queries at scale.
- **Visualisation**: the raylib visualiser places neurons at their real positions. HyperNEAT is related prior work.

Drift rules

- **Mutation**: a small Gaussian nudge, with an evolvable drift size so stable lineages settle down.

- **Recombination**: when both parents share a neuron, the child takes a random point between the two positions, avoiding strict averaging.
- **Activity-driven drift**: co-firing neurons pull together (Hebbian in space, like self-organising maps). Short-range repulsion or a minimum spacing prevents collapse.
- Moving neurons changes delays, so drift interacts with STDP.

Delay handling

Longer distances mean longer delays, so the spike-delivery logic must scale.

- **Ring buffer sized to max delay**: simple and fast; memory grows with the maximum.
- **Hybrid queue**: ring buffer for short local delays (most traffic), priority queue for rare long-range spikes.
- **Delay cap**: around 1–20 ms, as in Izhikevich's polychronization work.
- **Evolvable conduction speed per connection**: a myelination analogue, letting evolution tune timing without moving neurons.

Open questions

- Decided: 3D coordinates. 4D is an idea to revisit later.
- Does lifetime drift write back to the genome (Lamarckian) or not (Baldwinian)?

Suggested Build Order

1. A single column of Izhikevich neurons with in-column inhibition, working end to end.
2. Spike-window to SDR output.
3. Random receptive fields.
4. Teams and cooperative coevolution.
5. Cross-column inhibition, subpopulations, and diversity reward, each measured on its own contribution.

Later: GPU Execution

Once everything works, explore running the Izhikevich neuron updates on a GPU. The updates parallelise well, while spike routing with varied 3D delays is irregular and may stay on the CPU. Prior art: GeNN, CARLsim. Haskell route: the accelerate library. Benchmark against the many-core CPU plan.

Open Questions

- Should critters be columns at all, or should columns be fixed building blocks wired by NEAT?
- How should fitness be shared between a critter and its team?
- Which spike code to use as the primary scheme?